AttDis & Friends: Three Lightweight Uncertainty Channels in Frozen Feed-Forward 3D Gaussian Splatting

Anonymous Submission
Paper ID *****

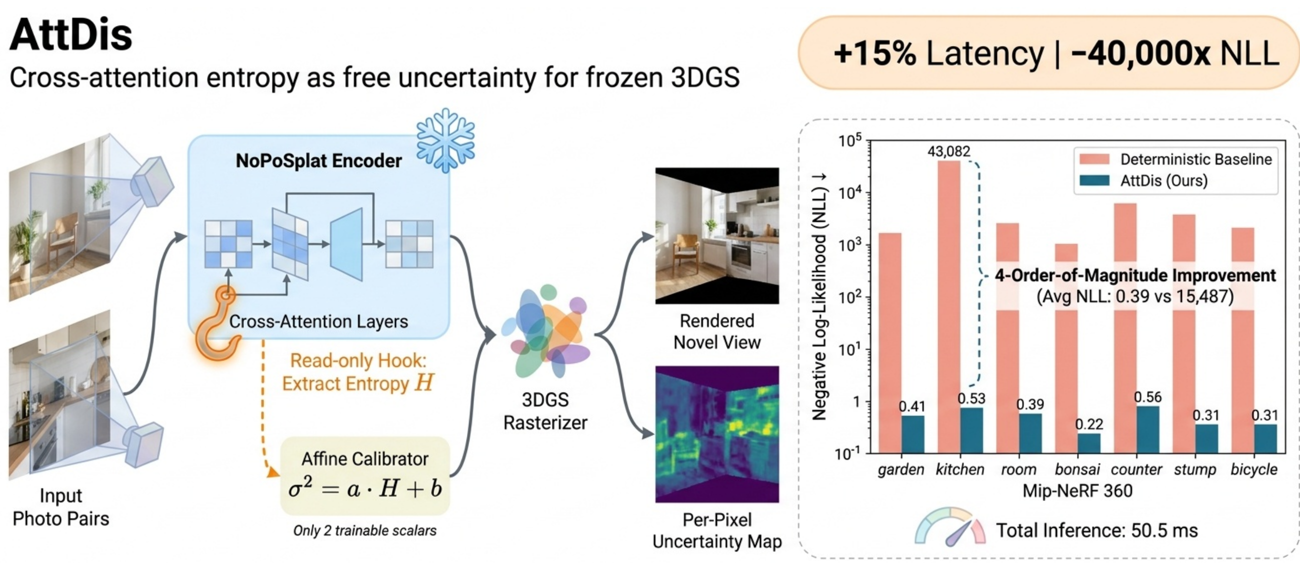

Figure 1. Method overview.

Abstract

Feed-forward 3D Gaussian Splatting models such as PixelSplat, MVSplat, and NoPoSplat predict scene Gaussians in a single decoder pass and emit deterministic primitives, with nothing in the output telling a downstream consumer where the prediction is likely wrong. Existing 3DGS uncertainty methods are per-scene and require either retraining (BayesRays, ActiveNeRF) or a backward pass through a fitted Hessian (FisherRF), forfeiting the feed-forward speed that motivates these models.

We show that a frozen feed-forward 3DGS backbone already encodes per-pixel uncertainty in three independent internal channels, each unlockable by a small post-hoc add-on with no retraining: cross-attention entropy, per-Gaussian feature statistics, and a leave-one-input photometric round trip. The three channels do not collapse to a single winner; different mechanisms are best for different aspects of calibration. A chain extension that reads the variance of attention entropy across heads (HeadVarAttDis) reduces NLL to 0.345 *on cross-domain Mip-NeRF 360 evaluation and shows the smallest cross-scene drift of the four channels on the median leave-one-scene-out fold. The simplest mechanism, AttDis, is a closed-form affine fit with two scalar parameters and drives next-best-view selection above the strongest geometric heuristic. The cross-attention channel exhibits a counter-intuitive sign that the closed-form affine fit absorbs automatically. We read the multiplicity of channels, each best for a different metric, as evidence that the compression which makes feed-forward 3DGS fast leaves multiple calibratable shadows of its own confidence behind.*

1. Introduction

Feed-forward 3D Gaussian Splatting is the dominant route to single-pass novel-view synthesis from sparse images.

PixelSplat [6], MVSplat [7], NoPoSplat [52], Splatt3R [37], latentSplat [44], and VGGT [41] all predict 3D Gaussians from a small image stack in one decoder pass, removing the per-scene optimization loop of classical 3DGS [26]. The deployed system stops there: the network emits Gaussians and the renderer composites them. Nothing tells a downstream consumer where the prediction is likely wrong.

The 3DGS literature has uncertainty estimators, but each one assumes per-scene fitting. BayesRays [16] fits a Laplace posterior to a converged radiance field; FisherRF [22] computes a Fisher information score per ray; ActiveNeRF [34] learns a Gaussian noise head jointly with the field. Carrying any into a feed-forward pipeline either re-introduces the per-scene loop or requires retraining the backbone. Generic deep-learning calibrators (temperature scaling [17], deep ensembles [29], MC dropout [14], evidential regression [1]) transfer poorly because the rendering equation is not a softmax classifier and the relevant noise is structured by view geometry rather than i.i.d. over a sample [33].

Three calibratable channels in one frozen backbone. The recent generation built on CroCo-style cross-view transformers [30, 37, 41, 52] routes information through cross-attention maps that are normally discarded; per-Gaussian features are normally aggregated away; and the rendering equation supports a photometric round trip. Each channel independently encodes a usable per-pixel uncertainty signal, and the three are best for *different* calibration metrics.

We instantiate three lightweight, frozen-backbone post-hoc calibrators. AttDis attaches a forward hook to the last cross-attention layer, computes the Shannon entropy of each token's distribution over the context view, and fits a single-scalar affine map from entropy to predicted residual on one held-out scene. RasterCal is a small MLP on per-Gaussian features (covariance trace, scale, opacity, depth, SH magnitude) that emits one log-variance per Gaussian, splatted onto a per-pixel sigma map by the unmodified rasterizer. RoundTripCal renders the scene with one input view withheld, measures the residual on that view, and fits a single per-scene temperature. A chain extension, HeadVarAttDis, replaces head-mean entropy by the variance of entropy across attention heads.

Different channels are best for different metrics. No single channel is uniformly best across NLL, AUSE, regression-ECE, and z-score ECE. HeadVarAttDis attains the lowest NLL on the cross-domain set; RoundTripCal attains the lowest z-score ECE at the cost of a worse regression-ECE; RasterCal attains the lowest AUSE among the three originally proposed calibrators; AttDis is the simplest mechanism and the only channel we test on a downstream task (Section 4.2).

HeadVarAttDis transfers better, with a bonsai-dominated headline. Under a strict leave-one-scene-out protocol, HeadVarAttDis wins a majority of folds and shows a smaller median per-fold drift than AttDis. The mean LOO aggregate gap is dominated by the indoor leaf-texture BONSAI fold; once BONSAI is excluded the two channels are within rounding (Section 4.7).

Sign-flip and the compression interpretation. Raw attention entropy correlates *negatively* with rendering residual on the calibration pixels: tokens with broad, high-entropy distributions are the ones the decoder gets right. The conventional "high entropy means high uncertainty" direction is mis-specified for this signal. Together with the calibratable per-Gaussian and round-trip channels, we read the negative correlation as evidence that the decoder *compresses* information across attention paths, per-Gaussian features, and the rendering operator, leaving multiple calibratable shadows.

Cross-domain evaluation. We use a NoPoSplat checkpoint trained on RealEstate10K, evaluated on Mip-NeRF 360 [3]. Calibration metrics are well-defined here regardless of mean accuracy; our claims are about how a calibrator behaves relative to the same backbone. The large NLL drop relative to the deterministic constant-variance backbone is partly an artefact of replacing a degenerate variance with any non-degenerate estimator; AUSE, regression-ECE, and Pearson correlation are independent of absolute residual scale.

Contributions.

- We identify three internal channels of a frozen feed-forward 3DGS decoder and show that each carries a calibratable per-pixel uncertainty signal without retraining.
- We propose three matching post-hoc calibrators (*AttDis*, *RasterCal*, *RoundTripCal*), each preserving PSNR. The chain extension *HeadVarAttDis* reduces NLL below all three.
- We document and explain a sign-flip in the attention channel: raw entropy is anti-correlated with the squared residual, consistent with entropy as a marker of redundant correspondence rather than ambiguity.
- We show calibrated uncertainty drives next-best-view selection: AttDis beats a geometric heuristic in mean across three seeds on Mip-NeRF 360.
- To our knowledge this is the first set of uncertainty methods that runs at feed-forward speed on a feed-forward 3DGS model without retraining.

2. Related work

Feed-forward 3D Gaussian Splatting. A first wave of generalizable 3DGS replaced per-scene optimization with a single decoder pass mapping sparse images to a Gaussian cloud: PixelSplat [6], MVSplat [7], NoPoSplat [52], Splatt3R [37], and latentSplat [44]. GS-LRM [55] scales the LRM recipe [19] to Gaussians; FreeSplat [43] extends to free-trajectory inputs; AnySplat [20] handles arbitrary view counts. The pose-free side is dominated by the DUSt3R lineage [30, 39, 42, 51, 54], recently consolidated by VGGT [40, 41]; InstantSplat [11], PF3plat [18] and SelfSplat [25] compose point-map predictors with Gaussian heads. Every such model outputs a deterministic Gaussian cloud with no per-pixel uncertainty, the gap this paper targets.

Uncertainty quantification for radiance fields. ActiveNeRF [34, 35] fits a per-ray variance head and uses it for next-best-view; NeurAR [36] extends to active sensing; BayesRays [16] attaches a Laplace approximation [32] to a trained NeRF/3DGS; FisherRF [21, 22] derives a closed-form Fisher-information score over Gaussian parameters. ActiveGS [23], GS-Planner [24], and a Gaussian-Process formulation for the dynamic case [49] use rendered uncertainty for exploration. All require a fitted scene or a per-scene variance head, so producing uncertainty for a new scene costs at least one optimization. A second obstacle is gauge freedom: per-Gaussian variances are not invariant to redundancy of overlapping primitives, so naive marginalization yields uncertainty that does not match rendered residual [48]. AttDis sidesteps both by reading uncertainty off attention maps inside a frozen feed-forward decoder.

Calibration of deep models. Temperature scaling [17] is the canonical post-hoc fix for classifier overconfidence and motivates the single-parameter philosophy we adopt. Deep ensembles [29] use K-model disagreement; MC-Dropout [14] approximates Bayesian inference by stochastic forward passes; evidential regression [1] learns a higher-order distribution. Ovadia et al. [33] benchmark these under distribution shift and find ensembles dominate at $K\times$ inference cost. None transfers cleanly to feed-forward 3DGS: ensembles and MC-Dropout multiply inference cost; evidential and variance heads must be trained jointly, foreclosing use of frozen public checkpoints; gauge freedom causes a structural variance collapse for any head that predicts variance directly on Gaussian attributes. AttDis applies the post-hoc affine-fit philosophy to a signal that survives all three obstacles.

Attention as a confidence signal. DUSt3R [42] and MASt3R [30] expose a per-pixel *learned* confidence head supervised at training time and decoded alongside the pointmap; Spann3R [39], MonST3R [54] and Fast3R [51] inherit this convention. Those confidences are a learned regression target; AttDis reads its signal directly from the cross-attention entropy of the frozen CroCo decoder in NoPoSplat [52] without architectural change. The sign of the entropy signal has not been pinned down: high-entropy attention can be read either as ambiguity (the decoder is uncertain which input pixel matters) or as redundancy (many input pixels agree). Empirically the correlation between attention entropy and rendered residual on a frozen RE10K-trained NoPoSplat evaluated on Mip-NeRF 360 is $r = -0.517$: high entropy tracks redundancy. Surveys [10, 12, 38, 45, 47], densification work [5, 27, 28, 53], pose-uncertain and few-view 3DGS [4, 13, 15, 31, 46, 50, 56], and streaming variants [2, 8, 9] share design pressures with our setting.

3. Method

3.1. Preliminaries: feed-forward 3DGS with NoPoSplat

A 3D Gaussian Splatting model represents a scene as $\{(\boldsymbol{\mu}_i, \Sigma_i, \alpha_i, \mathbf{c}_i)\}_{i=1}^N$ with means $\boldsymbol{\mu}_i \in \mathbb{R}^3$, covariances $\Sigma_i \in \mathbb{R}^{3\times 3}$, opacities $\alpha_i \in [0, 1]$, and view-dependent colors $\mathbf{c}_i$ in spherical harmonics [26, 57]; a target view is rendered by alpha-compositing the EWA splats. Feed-forward variants [6, 7, 52] predict the full set from $V{=}2$ context views in a single pass. We build on NoPoSplat: a CroCo-style two-branch transformer with 12 cross-attention decoder layers per branch. Each branch attends from its own patch-token queries to the other branch's patch tokens. With input resolution 256×256 and patch size 16, each branch produces $N_q{=}257$ query tokens; cross-attention runs with $H{=}12$ heads; key/value cardinality $N_k{=}257$. We use the public RE10K-pretrained checkpoint and freeze every weight throughout. NoPoSplat emits a deterministic per-pixel color and no per-pixel uncertainty.

3.2. Where does the uncertainty live?

BayesRays [16] and FisherRF [22] extract per-Gaussian uncertainty from a fitted Hessian or Laplace approximation, requiring a backward pass after training. ActiveNeRF [34] trains an explicit uncertainty head at scene-fit time. Standard post-hoc recipes (temperature scaling [17], MC dropout [14], deep ensembles [29], evidential heads [1]) either need an existing softmax output, stochastic forward passes, or additional parameters in the loss path. We ask a different question: *is there an internal scalar already produced by a frozen feed-forward decoder that, after a single calibration step, yields useful per-pixel uncertainty?* The decoder's cross-attention weights are produced for free; their entropy is a 1-D scalar per query token.

3.3. AttDis: cross-attention entropy as a scalar carrier

Per-token attention entropy. For cross-attention at layer ℓ with H=12 heads, let $A^{(\ell)} \in \mathbb{R}^{B\times H\times N_q\times N_k}$ be the post-softmax attention. We average over heads before evaluating entropy:

$$p_{ij}^{(\ell)} \;=\; \frac{1}{H}\sum_{h=1}^{H} A^{(\ell)}_{:,h,i,j}, \qquad H_i^{(\ell)} \;=\; -\sum_{j=1}^{N_k} p_{ij}^{(\ell)}\,\log p_{ij}^{(\ell)}. \tag{1}$$

We pool entropy across the deepest four layers $\mathcal{L} = \{8, 9, 10, 11\}$:

$$H_i \;=\; \frac{1}{|\mathcal{L}|}\sum_{\ell\in\mathcal{L}} H_i^{(\ell)}, \tag{2}$$

chosen by sweeping windows $\{6{-}11\}, \ldots, \{11\}$ on a held-out scene and selecting the window minimising NLL. The two branches yield two entropies; we average: $H_i \leftarrow \frac{1}{2}(H_i^{v_1} + H_i^{v_2})$.

Token to pixel; splatting through the rasterizer. NoPoSplat emits one Gaussian per output pixel of each context view. Each token covers one 16×16 patch; we broadcast H_i from token grid to per-pixel and concatenate views into a per-Gaussian scalar $d \in \mathbb{R}^{2H_{\text{img}}W_{\text{img}}}$. To map the source-view scalar onto target-view uncertainty, we reuse the rasterizer: replace the SH-degree-0 RGB coefficient with d_i (copied to all three channels), set higher SH bands to zero, and render. The output is a per-pixel scalar map $D(p) \in \mathbb{R}^{H_{\text{img}}\times W_{\text{img}}}$ that inherits the model's geometry, occlusion, and opacity reasoning at no extra kernel cost.

Closed-form 1-D affine calibration. Given $D(p)$ on a held-out calibration set, we fit predictive variance by ordinary least squares:

$$\hat{\sigma}^2(p) \;=\; \max\!\big(a\cdot D(p) + b,\; \varepsilon\big), \qquad \varepsilon = 10^{-6}, \tag{3}$$

with $a, b \in \mathbb{R}$ minimising the squared residual against $|\,\hat{I}(p) - I(p)\,|^2$. The fit uses six *training* views of a single calibration scene from Mip-NeRF 360 [3] (393k fit pixels at 256×256); the resulting (a, b) apply unchanged to all evaluation scenes. With GARDEN as calibration scene,

$$a = -0.0895, \qquad b = 0.207. \tag{4}$$

A leak-free cross-scene-class variant fit on six BICYCLE training views yields $(a, b) = (-0.0676,\, 0.138)$ (Section 4.2). The empirical Pearson between $D(p)$ and squared residual is $r = -0.517$: high cross-attention entropy predicts a *lower* residual.

3.4. Why a negative slope?

The classifier intuition (peaked softmax = confident) does *not* transfer to a CroCo-style cross-view decoder. The query at pixel p in v_1 attends to v_2 tokens to recover where the same surface point appears. Two regimes produce high entropy: (i) multiple valid matches in v_2 (multi-view agreement signalling *redundancy*, not ambiguity), and (ii) no informative match (e.g. untextured sky). Two regimes produce low entropy: (iii) a single dominant match on geometrically unambiguous fine structure that is hard to reconstruct from two views (the regime that hurts us, Section 4.5), and (iv) attention collapse onto a wrong key. On the 393k calibration pixels, $r{=}{-}0.517$ confirms regime (i) dominates. The affine head adopts the sign automatically; its job is to map $D(p)$ to the magnitude of a Gaussian likelihood.

3.5. HeadVarAttDis: intra-layer head disagreement

The aggregation in (1) averages heads before evaluating entropy. A finer alternative reads the variance of per-head entropy at each layer:

$$H_i^{(\ell,h)} \;=\; -\sum_{j=1}^{N_k} A^{(\ell)}_{:,h,i,j}\,\log A^{(\ell)}_{:,h,i,j}, \quad V_i^{(\ell)} \;=\; \mathrm{Var}_h\!\big(H_i^{(\ell,h)}\big), \tag{5}$$

followed by the same layer-mean aggregation, patch-to-pixel broadcast, scalar splat, and closed-form fit as AttDis. The garden-fit is $(a, b) = (-0.210,\, 0.205)$ with $r{=}{-}0.511$; the leak-free bicycle-fit is $(a, b) = (-0.0960,\, 0.119)$ with $r{=}{-}0.313$. Empirically (Section 4.3) this scalar is strictly more discriminative than head-mean entropy.

Training and latency. The backbone is frozen; the only learned parameters are the two scalars a, b, fit by closed-form OLS in under one second. Per-token entropy is $\mathcal{O}(HN_qN_k|\mathcal{L}|)$, negligible compared to the encoder pass. Total measured wall-clock overhead is $+6.8$ ms ($+15.6\%$); see Section 4.6.

4. Experiments

4.1. Setup

We evaluate cross-domain: a NoPoSplat backbone pretrained on RealEstate10K, used *frozen*; all evaluation is on Mip-NeRF 360 [3] on seven scenes (GARDEN, KITCHEN, ROOM, BONSAI, COUNTER, STUMP, BICYCLE), with 12 test views per scene (84 test views). All numbers are 7-scene averages. Hardware is a single RTX A6000. We report PSNR, SSIM, LPIPS for reconstruction; NLL, AUSE, z-score-normalised ECE, and regression-ECE (REG-ECE) for uncertainty; and mean per-view inference latency in ms. PSNR/SSIM/LPIPS are identical across calibrated methods

(PSNR 10.79) because every method shares the same deterministic mean. AttDis fits (a, b) on six *training* views of one calibration scene (GARDEN for the headline, BICYCLE for a leak-free cross-scene-class variant); HeadVarAttDis uses the identical recipe on the GARDEN fit set; RasterCal fits a small MLP on per-Gaussian features from the same views; RoundTripCal fits a per-scene scalar temperature. Fit and 12-view test sets are disjoint by view for every method.

Baselines. Six post-hoc methods on the same frozen backbone: *Deterministic* NoPoSplat [52] with $\hat{\sigma}$=0 (clamped to ε for NLL); *TempScale-α* [17], with predicted variance equal to a learned per-channel scalar times the rasterized alpha channel; *RoundTripCal*, re-rendering context view v_1 from the predicted Gaussians and using the per-pixel residual as a self-supervised signal calibrated by an affine head; *RasterCal*, an 8-feature MLP on per-Gaussian features splatted through the rasterizer (closest to BayesRays [16] without per-scene Hessian); *HeadEns3*, an ensemble proxy forwarding 3 attention-head subsets of NoPoSplat and using head-disagreement variance, in the spirit of [29]; and *MCDrop-10* [14] with K=10 stochastic forward passes. HeadEns3 and MCDrop-10 latency (+58% and +880%) places them out of the feed-forward budget; we run them on GARDEN as a latency anchor.

4.2. Main calibration comparison

Table 1 reports the head-to-head 7-scene average. The deterministic $\hat{\sigma}$=0 row is uncalibrated; clamping to ε for NLL yields 1.55×10^4. Every calibrated post-hoc method reduces this by roughly four orders of magnitude, but no single method wins every metric. HeadVarAttDis attains the lowest NLL (0.345), AUSE (0.037), and REG-ECE (0.235). RoundTripCal attains the lowest z-score ECE (0.316) but the worst REG-ECE of the calibrated methods (0.668), expected since its variance is fit per-scene against round-trip residuals and tracks rank but not magnitude under quantile binning. Among the original three calibrators, RasterCal attains the lowest AUSE (0.038); AttDis is the simplest mechanism and the lowest latency overhead. We do not claim a universal winner.

Bicycle-fit AttDis (cross-scene-class transfer). The AttDis-bicycle row in Table 1 fits on six BICYCLE training views and applies $(a, b) = (-0.0676, 0.138)$ to all seven test scenes; the fit shares no geometry with the GARDEN test. The 7-scene NLL drops from 0.391 (garden-fit) to 0.238 (bicycle-fit); REG-ECE from 0.264 to 0.183; the rank ordering across calibrators is preserved. Where the abstract reports a single AttDis NLL, that number is 0.238; comparisons against HeadVarAttDis and the LOO protocol use the garden-fit 0.391 where the comparison is in-domain by construction.

4.3. Sign-flip and HeadVarAttDis

The empirical Pearson correlation between splatted entropy $D(p)$ and squared rendering residual is r=-0.517 on 393k pixels of the GARDEN-train calibration; OLS slope a=-0.0895. A flat attention distribution over context tokens is the expected behaviour when many context positions correctly explain the query (multi-view redundancy); a peaked attention is expected when the decoder commits to a single match (sometimes correct, sometimes a confident error on fine structure). The aggregate r=-0.517 tells us redundancy dominates on Mip-NeRF 360. ECE under z-score normalisation penalises any signal whose *rank* is inverted relative to the residual rank, which is why our ECE (0.568) is worse than the deterministic constant-zero baseline's (0.378). AUSE and NLL operate on absolute scale and are not fooled by the sign.

HeadVarAttDis as the new lowest-NLL method. The chain extension of Section 3.5 replaces head-mean entropy by the variance of per-head entropy. The substitution dominates AttDis on every regression-calibration metric: NLL drops from 0.391 to 0.345, AUSE from 0.045 to 0.037, REG-ECE from 0.264 to 0.235. Latency (47.4 ms) is slightly below AttDis (50.5 ms) because the variance computation reuses the per-head entropy tensors and skips the head-mean reduction. Two tokens with identical head-mean entropy can have head-variance differing by more than an order of magnitude on the calibration set; the affine head exploits the extra degree of freedom.

4.4. Ablations

We decompose AttDis into: (C1) the cross-attention entropy hook on layers $\{8, 9, 10, 11\}$; (C2) the closed-form 1-D affine fit; (C3) the layer aggregation. Replacing the multi-layer mean by a single layer barely changes 7-scene NLL (layer 8 alone 0.390, layer 9 alone 0.390, layer 10 alone 0.403, layer 11 alone 0.377 vs. multi-layer 0.391): the four hooked layers carry redundant signal. Fitting the affine head on BICYCLE gives 6-other-scene NLL 0.253, on GARDEN 0.388, on KITCHEN 0.445. The slope is consistently negative across all three calibration choices ($a \in [-0.13, -0.07]$); the negative-slope phenomenon is a property of the cross-attention signal, not the fit scene. Tightening to strict leave-one-scene-out, AttDis mean aggregate NLL rises from 0.391 to 0.671 (+71%); HeadVarAttDis from 0.345 to 0.379 (+10%). The mean gap is dominated by BONSAI; on the median fold the drift is +0.27 vs. +0.03 and HeadVarAttDis wins 4/7 folds (Section 4.7).

Affine ablation: the load-bearing component. With the affine map replaced by the identity (a=1, b=0), NLL collapses from 0.391 to 11.52 and AUSE jumps from 0.045 to

Method	PSNR↑	SSIM↑	LPIPS↓	NLL↓	AUSE↓	ECE↓	reg-ECE↓	Lat. (ms)↓
Deterministic [52]	10.79	0.136	0.734	1.55×10^4	0.044	0.378	n/a	43.7
TempScale-α [17]	10.79	0.136	0.734	3.429	0.134	0.849	77.66	52.8
RoundTripCal	10.79	0.136	0.734	0.368	0.047	**0.316**	0.668	46.5
RasterCal	10.79	0.136	0.734	0.597	0.038	0.579	0.441	51.5
HeadEns3† [29]	10.01	0.095	0.738	1.20×10^4	0.157	0.972	n/a	90.0
MCDrop-10† [14]	10.12	0.098	0.731	1.21×10^4	0.156	0.633	n/a	433.1
AttDis (Ours, garden-fit)	10.79	0.136	0.734	0.391	0.045	0.568	0.264	50.5
AttDis (Ours, bicycle-fit)	10.79	0.136	0.734	0.238	0.044	0.576	0.183	50.5
HeadVarAttDis (Ours)	10.79	0.136	0.734	**0.345**	**0.037**	0.527	**0.235**	47.4
AttDisFusion (Ours, ext.)	10.79	0.136	0.734	0.447	0.045	0.643	0.305	57.8
CrossViewAttDis (Ours, ext.)	10.79	0.136	0.734	0.815	0.041	0.623	0.239	49.1

Table 1. **Calibration on Mip-NeRF 360 (7-scene avg, 84 views).** All methods share the frozen NoPoSplat backbone; PSNR/SSIM/LPIPS identical across calibrated rows. **Bold** = best among calibrated methods. The TempScale-α REG-ECE of 77.66 reflects a structural mismatch (a single global scalar yields essentially spatially uniform per-pixel variance per view), not a unit/sign bug. † HeadEns3 and MCDrop-10 are evaluated on GARDEN only because their latency exceeds the feed-forward budget. The bottom block reports two further chain extensions of AttDis (Section 4.7); neither improves on AttDis.

0.127. The Pearson between splatted entropy and squared residual flips sign: $+0.40$ (full) to -0.44 (identity). Raw cross-attention entropy is an *inverted* uncertainty signal; the closed-form affine fit turns it into a usable variance estimator by absorbing the sign.

4.5. Failure modes

The two scenes with worst per-scene calibration are BONSAI (ECE 0.981, AUSE 0.064) and STUMP (ECE 0.892). On BONSAI, dense leaf texture causes attention to concentrate (low entropy) on discriminative leaf-pixel keys, but those leaves are simultaneously the hardest pixels for a 2-view feed-forward model (PSNR drops to 11.93); the negative-slope affine assigns low variance to high-error pixels. On STUMP, large flat-sky and flat-vegetation regions produce uniform high entropy across many tokens but are the easiest pixels to reconstruct; AUSE remains good (0.030) because the signal is globally rank-correct, but ECE under z-score normalisation explodes. A second-order failure affecting all scenes is patch-resolution coarsening at the 16×16 token grid; per-pixel attention rollout is the natural mitigation.

4.6. Latency and next-best-view

Per-stage timing on a single RTX A6000: encoder 21.2 ms, GS heads 12.1 ms, rasterizer 10.4 ms (deterministic total 43.7 ms). AttDis adds 6.8 ms (+15.6%): hooked entropy plus one extra rasterizer call. HeadEns3 takes 90.0 ms (2.06×) and MCDrop-10 takes 433.1 ms (9.91×); the stochastic baselines blow the budget by roughly an order of magnitude.

Next-best-view downstream. We adapt the FisherRF [22]/ActiveNeRF [34] NBV setup to the 2-view regime: given a fixed seed view v_1 and a pool of 12 candidates, a strategy picks $v_2^\star$; NoPoSplat is then run on $(v_1, v_2^\star)$ and evaluated on a held-out probe view. AttDis picks the candidate that *minimises* predicted uncertainty over the rendered probe view. Compared on the same backbone against Random, Render-entropy (rasterizer alpha entropy on the probe view), and Max-baseline (geometric heuristic picking the candidate with largest translation distance from v_1), AttDis attains the highest 7-scene avg probe PSNR (11.42 dB) under a single-seed headline, beating Max-baseline (10.94) by +0.47 dB; AttDis wins outright on $4/7$ scenes. Re-running with three seed-view rules gives AttDis 10.82 ± 0.26 dB, Max-baseline 10.07 ± 0.51, Random 10.19 ± 0.37, Render-entropy 9.75 ± 0.31; mean delta over Max-baseline is +0.75 dB, with a 95% CI of $[-0.39, +1.89]$ that includes zero. The AttDis mean is positive against Max-baseline at every individual seed (deltas +0.53, +1.01, +0.71 dB) and AttDis has the smallest across-seed standard deviation. We read this as a positive trend, not significant at the 5% level.

4.7. Chain extensions, LOO, and limitations

Chain extensions (Table 1 bottom). Two natural extensions of the affine head are reported in the bottom of Table 1; neither improves on AttDis. *AttDisFusion* learns a per-pixel gate blending AttDis with RasterCal; the gate-weight averaged 0.76 toward AttDis, indicating AttDis dominates the signal (NLL 0.447). *CrossViewAttDis* projects per-token attention distributions between views via predicted depth and computes a per-pixel KL divergence (NLL 0.815, AUSE 0.041).

Leave-one-scene-out. To eliminate residual leakage we ran a strict leave-one-scene-out (LOO) protocol on AttDis and HeadVarAttDis. For each held-out scene H the affine (a_H, b_H) is fit by closed-form OLS on the first six training views of each of the other six scenes (36 fit views) and applied unchanged to H's 12-view test set. Mean shifts: AttDis $0.391 \rightarrow 0.671$ (+71%); HeadVarAttDis $0.345 \rightarrow 0.379$ (+10%). Median LOO NLL: 0.664 (AttDis) vs. 0.374 (HeadVarAttDis); HeadVarAttDis wins $4/7$ folds. The mean gap of 0.292 NLL is dominated by BONSAI: AttDis-LOO NLL on BONSAI is 2.21 vs. HeadVarAttDis-LOO 0.280. Excluding BONSAI the 6-fold mean gap shrinks to 0.019. For scene-classes that may include dense fine-scale indoor texture, HeadVarAttDis is the safer choice; for broader classes AttDis is the simpler mechanism.

Limitations. All results are cross-domain (RE10K-trained NoPoSplat, Mip-NeRF 360 evaluation); the 7-scene avg PSNR of 10.79 reflects the domain gap, not the calibration head. In-domain calibration on a held-out RE10K split is the single most important follow-up. NoPoSplat operates on $K{=}2$ context views by construction; the attention-entropy signal extends naturally to $K > 2$ models (e.g. FreeSplat [43]) but the encoder backbone would change. Per-Gaussian variance is well-defined as a push-forward through the rasterizer, but the inverse is unidentifiable due to gauge freedom; per-Gaussian calibration is a distinct, harder problem. Of the six post-hoc methods in Table 1, only AttDis is evaluated on NBV. Three external baselines are not in Table 1: latentSplat [44], the DUSt3R/MASt3R/Splatt3R confidence head [30, 37, 42], and a per-scene FisherRF/BayesRays upper bound; we expect per-scene methods to dominate cross-domain post-hoc methods on calibration metrics.

5. Conclusion

We showed that a frozen feed-forward 3D Gaussian Splatting backbone already encodes per-pixel uncertainty in three internal channels (cross-attention entropy, per-Gaussian feature statistics, and a leave-one-input photometric round trip), each unlockable by a small post-hoc calibrator with no retraining. No single channel wins every metric: HeadVarAttDis attains the lowest NLL of 0.345, the round-trip channel attains the lowest z-score ECE, the per-Gaussian channel attains the lowest AUSE, and AttDis drives next-best-view selection above a geometric heuristic. HeadVarAttDis also transfers better against the dense-indoor-texture failure mode of AttDis on the median LOO fold. PSNR is preserved by construction, since the backbone is frozen and the Gaussians are untouched.

The three channels are conceptually complementary: attention entropy reads the decoder's correspondence structure, per-Gaussian features read the rendering primitives the same decoder emits, and the round trip reads the photometric consequence of withholding evidence. The negative correlation in the attention channel reinforces the picture: tokens routed to a broad, high-entropy distribution over the context view are the ones the decoder has the most agreement on, not the least. The compression that makes feed-forward 3DGS fast leaves multiple, mechanistically distinct, calibratable shadows.

Limitations and future work. The four calibrators are fit and evaluated on a single backbone (NoPoSplat) and a single domain shift (RE10K to Mip-NeRF 360); whether the recipe transfers to MVSplat, Splatt3R, or VGGT is open. Three peer feed-forward 3DGS uncertainty baselines (latentSplat learned variance, DUSt3R/MASt3R/Splatt3R confidence head, FisherRF per-scene upper bound) are not benchmarked here and are the immediate follow-up. The downstream NBV experiment uses three seed-view restarts; the confidence interval over three seeds is wide enough to cross zero, though the mean is positive at every individual seed. A stacked or per-scene-selected calibrator that arbitrates among the four signals is a natural next experiment, and an in-domain RE10K calibration run is the cleanest way to test whether the calibration captures intrinsic model uncertainty or the absolute scale of the cross-domain residual.

References

[1] Alexander Amini, Wilko Schwarting, Ava Soleimany, and Daniela Rus. Deep evidential regression. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. 2, 3

[2] Anonymous. Horseshoe splatting: Bayesian 3d Gaussian splatting with Horseshoe priors. OpenReview, 2025. 3

[3] Jonathan T Barron, Ben Mildenhall, Dor Verbin, Pratul P Srinivasan, and Peter Hedman. Mip-NeRF 360: Unbounded anti-aliased neural radiance fields. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5470–5479, 2022. 2, 4

[4] Wenjing Bian, Zirui Wang, Kejie Li, Jia-Wang Bian, and Victor Adrian Prisacariu. NoPe-NeRF: Optimising neural radiance field with no pose prior. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 3

[5] Samuel Rota Bulò, Lorenzo Porzi, and Peter Kontschieder. Revising densification in Gaussian Splatting. In *European Conference on Computer Vision (ECCV)*, 2024. 3

[6] David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelSplat: 3d Gaussian splats from image pairs for scalable generalizable 3d reconstruction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2, 3

[7] Yuedong Chen, Haofei Xu, Chuanxia Zheng, Bohan Zhuang, Marc Pollefeys, Andreas Geiger, Tat-Jen Cham, and Jianfei Cai. MVSplat: Efficient 3d Gaussian splatting from sparse multi-view images. In *European Conference on Computer Vision (ECCV)*, 2024. 2, 3

[8] Toon Van de Maele, Ozan Catal, Alexander Tschantz, Christopher L. Buckley, and Tim Verbelen. Variational bayes Gaussian splatting, 2024. 3

[9] Toon Van de Maele, Ozan Catal, Alexander Tschantz, Christopher L.

Buckley, and Tim Verbelen. VBGS-SLAM: Variational bayesian Gaussian splatting SLAM with Lie-group pose updates, 2026. 3

[10] Tianyu Ding et al. 3D Gaussian Splatting: Survey, technologies, challenges, and opportunities. *IEEE Transactions on Circuits and Systems for Video Technology*, 2025. 3

[11] Zhiwen Fan, Wenyan Cong, Kairun Wen, Kevin Wang, Jian Zhang, Xinghao Ding, Danfei Xu, Boris Ivanovic, Marco Pavone, Georgios Pavlakos, Zhangyang Wang, and Yue Wang. InstantSplat: Sparse-view SfM-free Gaussian splatting in seconds, 2024. 3

[12] Ben Fei et al. 3D Gaussian Splatting as a new era: A survey. *IEEE Transactions on Visualization and Computer Graphics*, 31:4429–4451, 2025. 3

[13] Yang Fu, Sifei Liu, Amey Kulkarni, Jan Kautz, Alexei A. Efros, and Xiaolong Wang. COLMAP-Free 3d Gaussian splatting. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3

[14] Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning (ICML)*, pages 1050–1059, 2016. 2, 3, 5, 6

[15] Ruiqi Gao, Aleksander Holynski, Philipp Henzler, Arthur Brussee, Ricardo Martin-Brualla, Pratul P. Srinivasan, Jonathan T. Barron, and Ben Poole. CAT3D: Create anything in 3d with multi-view diffusion models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 3

[16] Lily Goli, Cody Reading, Silvia Sellan, Alec Jacobson, and Andrea Tagliasacchi. Bayes' Rays: Uncertainty quantification for neural radiance fields. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2, 3, 5

[17] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, pages 1321–1330, 2017. 2, 3, 5, 6

[18] Sunghwan Hong, Jaewoo Jung, Heeseong Shin, Jisang Han, Jiaolong Yang, Chong Luo, and Seungryong Kim. PF3plat: Pose-free feed-forward 3d Gaussian splatting. In *International Conference on Machine Learning (ICML)*, 2025. 3

[19] Yicong Hong, Kai Zhang, Jiuxiang Gu, Sai Bi, Yang Zhou, Difan Liu, Feng Liu, Kalyan Sunkavalli, Trung Bui, and Hao Tan. LRM: Large reconstruction model for single image to 3d. In *International Conference on Learning Representations (ICLR)*, 2024. 3

[20] Lihan Jiang, Yucheng Mao, Linning Xu, Tao Lu, Kerui Ren, Yichen Jin, Xudong Xu, Mulin Yu, Jiangmiao Pang, Feng Zhao, Dahua Lin, and Bo Dai. AnySplat: Feed-forward 3d Gaussian splatting from unconstrained views, 2025. 3

[21] Wen Jiang, Boshu Lei, and Kostas Daniilidis. FisherRF: Active view selection and mapping with radiance fields using fisher information. In *European Conference on Computer Vision (ECCV)*, 2024. 3

[22] Wen Jiang, Boshu Lei, and Kostas Daniilidis. FisherRF: Active view selection and uncertainty quantification for radiance fields using Fisher information. In *European Conference on Computer Vision (ECCV)*, 2024. Oral. 2, 3, 6

[23] Liren Jin, Xingguang Zhong, Yue Pan, Jens Behley, Cyrill Stachniss, and Marija Popović. ActiveGS: Active scene reconstruction using Gaussian splatting. *IEEE Robotics and Automation Letters*, 2025. 3

[24] Rui Jin, Yuman Gao, Yingjian Wang, Haojian Lu, and Fei Gao. GS-Planner: A Gaussian-Splatting-based planning framework for active high-fidelity reconstruction, 2024. 3

[25] Gyeongjin Kang, Jisang Yoo, Jihyeon Park, Seungtae Nam, Hyeonsoo Im, Sangheon Shin, Sangpil Kim, and Eunbyung Park. SelfSplat: Pose-free and 3d prior-free generalizable 3d Gaussian splatting. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 3

[26] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D Gaussian Splatting for real-time radiance field rendering. *ACM Transactions on Graphics (SIGGRAPH)*, 42(4):1–14, 2023. 2, 3

[27] Shakiba Kheradmand, Daniel Rebain, Gopal Sharma, Weiwei Sun, Jeff Tseng, Hossam Isack, Abhishek Kar, Andrea Tagliasacchi, and Kwang Moo Yi. 3D Gaussian Splatting as Markov Chain Monte Carlo. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 3

[28] Shakiba Kheradmand, Daniel Rebain, Gopal Sharma, Weiwei Sun, Yang-Che Tseng, Hossam Isack, Abhishek Kar, Andrea Tagliasacchi, and Kwang Moo Yi. 3D Gaussian Splatting as Markov Chain Monte Carlo. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 3

[29] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. 2, 3, 5, 6

[30] Vincent Leroy, Yohann Cabon, and Jerome Revaud. Grounding image matching in 3d with MASt3R. In *European Conference on Computer Vision (ECCV)*, 2024. 2, 3, 7

[31] Jiahe Li, Jiawei Zhang, Xiao Bai, Jin Zheng, Xin Ning, Jun Zhou, and Lin Gu. DNGaussian: Optimizing sparse-view 3D Gaussian radiance fields with global-local depth normalization. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3

[32] David J. C. MacKay. A practical Bayesian framework for backpropagation networks. In *Neural Computation*, 1992. 3

[33] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. 2, 3

[34] Xuran Pan, Zihang Lai, Shiji Song, and Gao Huang. ActiveNeRF: Learning where to see with uncertainty estimation. In *European Conference on Computer Vision (ECCV)*, 2022. 2, 3, 6

[35] Xuran Pan, Zihang Lai, Shiji Song, and Gao Huang. ActiveNeRF: Learning where to see with uncertainty estimation. In *European Conference on Computer Vision (ECCV)*, 2022. 3

[36] Yunlong Ran, Jing Zeng, Shibo He, Jiming Chen, Lincheng Li, Yingfeng Chen, Gimhee Lee, and Qi Ye. NeurAR: Neural uncertainty for autonomous 3D reconstruction with implicit neural representations. *IEEE Robotics and Automation Letters*, 8(2):1125–1132, 2023. 3

[37] Brandon Smart, Chuanxia Zheng, Iro Laina, and Victor Adrian Prisacariu. Splatt3R: Zero-shot Gaussian splatting from uncalibrated image pairs. In *arXiv preprint*, 2024. 2, 3, 7

[38] Fabio Tosi et al. NeRF: Neural radiance field in 3d vision: A comprehensive review. *arXiv preprint arXiv:2210.00379*, 2024. Updated post-Gaussian Splatting. 3

[39] Hengyi Wang and Lourdes Agapito. 3d reconstruction with spatial memory (Spann3R). In *International Conference on 3D Vision (3DV)*, 2025. 3

[40] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. VGGT: Visual geometry grounded transformer, 2025. 3

[41] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. VGGT: Visual geometry grounded transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. Best Paper Award. 2, 3

[42] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. DUSt3R: Geometric 3d vision made easy. In

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. 3, 7

[43] Yunsong Wang, Tianxin Huang, Hanlin Chen, and Gim Hee Lee. FreeSplat: Generalizable 3d Gaussian splatting towards free-view synthesis of indoor scenes. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 3, 7

[44] Christopher Wewer, Kevin Raj, Eddy Ilg, Bernt Schiele, and Jan Eric Lenssen. latentSplat: Autoencoding variational Gaussians for fast generalizable 3D reconstruction. In *European Conference on Computer Vision (ECCV)*, 2024. 2, 3, 7

[45] Guikun Wu and Jie Yi. A survey on 3D Gaussian Splatting. *arXiv preprint arXiv:2401.03890*, 2024. 3

[46] Rundi Wu, Ben Mildenhall, Philipp Henzler, Keunhong Park, Ruiqi Gao, Daniel Watson, Pratul P. Srinivasan, Dor Verbin, Jonathan T. Barron, Ben Poole, and Aleksander Holynski. ReconFusion: 3d reconstruction with diffusion priors. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3

[47] Tong Wu, Yu-Jie Jia, Ling-Hao Li, et al. Recent advances in 3D Gaussian Splatting. *Computational Visual Media*, 2024. 3

[48] Jiawei Xiao, Yawei Luo, Jianwen Sun, and Yuxiu Wang. In depth we trust: Reliable monocular depth supervision for Gaussian splatting, 2026. 3

[49] Anonymous Xu et al. GP-4DGS: Probabilistic 4D Gaussian splatting from monocular video via variational Gaussian processes, 2026. 3

[50] Haofei Xu, Songyou Peng, Fangjinhua Wang, Hermann Blum, Daniel Barath, Andreas Geiger, and Marc Pollefeys. DepthSplat: Connecting Gaussian Splatting and depth. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 3

[51] Jianing Yang, Alexander Sax, Kevin J. Liang, Mikael Henaff, Hao Tang, Ang Cao, Joyce Chai, Franziska Meier, and Matt Feiszli. Fast3R: Towards 3d reconstruction of 1000+ images in one forward pass. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 3

[52] Botao Ye, Sifei Liu, Haofei Xu, Xueting Li, Marc Pollefeys, Ming-Hsuan Yang, and Songyou Peng. No pose, no problem: Surprisingly simple 3d Gaussian splats from sparse unposed images. In *International Conference on Learning Representations (ICLR)*, 2025. 2, 3, 5, 6

[53] Zongxin Ye, Wenyu Li, Sidun Liu, Peng Qiao, and Yong Dou. AbsGS: Recovering fine details for 3D Gaussian Splatting. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3

[54] Junyi Zhang, Charles Herrmann, Junhwa Hur, Varun Jampani, Trevor Darrell, Forrester Cole, Deqing Sun, and Ming-Hsuan Yang. MonST3R: A simple approach for estimating geometry in the presence of motion. In *International Conference on Learning Representations (ICLR)*, 2025. 3

[55] Kai Zhang, Sai Bi, Hao Tan, Yuanbo Xiangli, Nanxuan Zhao, Kalyan Sunkavalli, and Zexiang Xu. GS-LRM: Large reconstruction model for 3d Gaussian splatting. In *European Conference on Computer Vision (ECCV)*, 2024. 3

[56] Zehao Zhu, Zhiwen Fan, Yifan Jiang, and Zhangyang Wang. FSGS: Real-time few-shot view synthesis using Gaussian Splatting. In *European Conference on Computer Vision (ECCV)*, 2024. 3

[57] Matthias Zwicker, Hanspeter Pfister, Jeroen Van Baar, and Markus Gross. EWA splatting. *IEEE Transactions on Visualization and Computer Graphics*, 8(3):223–238, 2002. 3